User Space vs Kernel space

Q1

Which of the following is a key advantage of user-space networking compared to kernel-space networking?

- All of the above
- Faster development and deployment without requiring kernel recompilation
- Reduced latency by bypassing the kernel
- Allows zero copy networking by directly sharing userspace buffers with hardware

Answer: All of the above

User-space networking has several advantages over kernel-space networking:

1. **Faster development and deployment without requiring kernel recompilation**
    - Since user-space networking bypasses kernel modifications, developers can update and optimize their networking stack without needing to modify or recompile the kernel.
    - This is particularly useful for rapid prototyping and deploying new networking protocols without requiring administrative privileges or waiting for kernel updates.
2. **Reduced latency by bypassing the kernel**
    - Traditional kernel networking involves system calls and context switching between user space and kernel space, which adds latency.
    - User-space networking frameworks (such as DPDK, eBPF/XDP, and netmap) avoid these context switches, reducing overhead and achieving lower latency.

3. **Allows zero-copy networking by directly sharing user-space buffers with hardware**
    - In kernel networking, packets are typically copied between user space and kernel space, introducing additional memory operations.
    - User-space networking frameworks enable zero-copy techniques, allowing direct memory access (DMA) between user-space buffers and the NIC, significantly improving performance.

Thus, the correct answer is:
"All of the above"

Pipelining and Performance

Q2

A network packet processing system consists of 3 stages: Parsing and Classification, Encryption and Routing. The Encryption stage is parallelized across 2 processors, while the other stages operate sequentially. The latencies of the stages are as follows:

Parsing and Classification: 4 cycles

Encryption (parallelized): 8 cycles per processor

Routing: 4 cycles

What is the latency for each packet to pass through the entire pipeline?

- 16 cycles
- 20 cycles
- 32 cycles

- 64 cycles

Answer: 16 cycles

We analyze the stages step by step:

- **Parsing and Classification**: 4 cycles (sequential)
- **Encryption**: This stage is parallelized across 2 processors, each taking 8 cycles. Since both processors can work in parallel, the effective encryption latency is still **8 cycles** (not 16).
- **Routing**: 4 cycles (sequential)

Total latency per packet:
4 (Parsing & Classification) + 8 (Encryption) + 4 (Routing) = 16 cycles

Thus, the correct answer is:
"16 cycles"

Q3

A network packet processing pipeline has 4 stages: Classification, Encryption, Routing, and Forwarding. Each stage takes 10 cycles to process a packet. If the pipeline is fully loaded and no delays occur between stages, how many packets can be processed per cycle?

- 1 packet per cycle
- 4 packets per cycle
- 1 packet every 4 cycles
- 1 packet every 10 cycles

1. In a **non-pipelined system**, each packet would take **4 × 10 = 40 cycles** to complete.
2. However, in a **pipelined system**, once the pipeline is filled, a new packet can enter the system **every** 10 **cycle**s and advance to the next stage.

3. This means **one packet completes every 10 cycles**

Throughput Calculation:

- Since a fully loaded pipeline processes **one packet per 10 cycles**, the correct answer is:

  1 Packet per 10 cycles

Q4

A network packet processing system consists of 3 stages: Parsing and Classification, Encryption and Routing. The Encryption stage is parallelized across 2 processors, while the other stages operate sequentially. The latencies of the stages are as follows:

Parsing and Classification: 4 cycles

Encryption (parallelized): 8 cycles per processor

Routing: 4 cycles

After the pipeline is filled, how many cycles will it take to process 3 packets?

- 20 cycles
- 24 cycles
- 32 cycles
- 36 cycles

Answer: 24 cycles

The pipeline works as follows:

1. **Packet 1:**
   - Cycle 1–4: Parsing & Classification
   - Cycle 5–12: Encryption (parallelized)
   - Cycle 13–16: Routing
   - **Packet 1 completes at cycle 16**
2. **Packet 2 enters while Packet 1 is still processing:**
   - Cycle 5–8: Parsing & Classification
   - Cycle 9–16: Encryption
   - Cycle 17–20: Routing
   - **Packet 2 completes at cycle 20**
3. **Packet 3 enters:**
   - Cycle 9–12: Parsing & Classification
   - Cycle 13–20: Encryption
   - Cycle 21–24: Routing
   - **Packet 3 completes at cycle 24**

Since we are asked how many cycles it takes to process 3 packets **after the pipeline is filled**, we calculate from the start of Packet 1 to the completion of Packet 3.

Total cycles needed: **24 cycles**

Thus, the correct answer is:
"24 cycles"

Q5

Using Amdahl's Law, if 80% of a task can be parallelized and the remaining 20% must run sequentially, what is the maximum speedup achievable when using 4 processors?

- 3.5
- 2.5
- 4.5
- 5

Answer: 2.5

Amdahl's Law is given by:

$$S = \frac{1}{(1 - P) + \frac{P}{N}}$$

where:

- S = speedup
- P = parallelizable portion of the task (0.8 or 80%)
- N = number of processors (4)
- 1−P = sequential portion of the task (0.2 or 20%)

Substituting the values:

$$S = \frac{1}{(0.2) + \frac{0.8}{4}}$$

$$S = \frac{1}{0.2 + 0.2} = \frac{1}{0.4} = 2.5$$

Thus, the correct answer is:
"2.5"

VLAN

Q6

Which technology enables the creation of multiple virtual networks over a shared physical network infrastructure?

- Network Address Translation (NAT)
- Multiprotocol Label Switching (MPLS)
- Virtual LAN (VLAN)
- Software Defined Networking (SDN)

Answer is **Virtual LAN (VLAN)**.

Here's why:

- **VLANs (Virtual LANs)** allow you to segment a physical network into multiple logical networks. Each VLAN acts like its own separate network, even though they all share the same underlying hardware. This is exactly what the question describes.

Let's look at why the other options aren't the best fit:

- **Software Defined Networking (SDN)** is an architectural approach to network management that allows network administrators to manage network services and resources through software. While SDN can *utilize* VLANs, it's not the technology *itself* that creates them. SDN is more about how the network is controlled.

- **Network Address Translation (NAT)** is used to translate private IP addresses into public IP addresses, allowing multiple devices on a private network to share a single public IP address. It doesn't create virtual networks.

- **Multiprotocol Label Switching (MPLS)** is a technique for forwarding data packets in a network. It's often used by service providers to create virtual private networks (VPNs) and for traffic engineering, but it's not the core technology for creating VLANs on a local network.

Q7

Which of the following is NOT a characteristic of VLAN (Virtual Local Area Network)?

- VLANs reduce broadcast domains
- VLANs can segment traffic within the same physical switch
- VLANs increase the number of collision domains
- VLANs work only with layer 3 switches

Answer: VLANs work only with layer 3 switches

Explanation:

- **VLANs operate at Layer 2 (Data Link Layer)** but can also be used in Layer 3 switches.
- **They reduce broadcast domains** by logically segmenting networks.
- **They can segment traffic within the same physical switch**, allowing better network organization.
- **They increase the number of collision domains** because each VLAN acts like an independent switch.

Correct Answer: VLANs work only with layer 3 switches (This is false, as VLANs work with Layer 2 switches as well.)

TCP Connection management

Q8

Which of the following is true about the TCP three-way handshake?

- It involves the exchange of a sequence number, acknowledgment number, and a checksum
- It establishes a reliable connection by transferring data in a single packet
- It is used to determine the maximum window size
- It uses a sliding window mechanism for flow control

Answer: It involves the exchange of a sequence number, acknowledgment number, and a checksum

The TCP three-way handshake consists of three steps:

1. **SYN**: The client sends a **SYN** packet with an initial sequence number.
2. **SYN-ACK**: The server responds with **SYN-ACK**, acknowledging the client's sequence number and sending its own sequence number.
3. **ACK**: The client responds with an **ACK**, completing the handshake.

During this process, sequence numbers, acknowledgment numbers, and checksums are exchanged to establish a reliable connection.

Explanation of other options:

- **"It establishes a reliable connection by transferring data in a single packet"** ❌ Incorrect – Data transfer happens **after** the handshake.
- **"It is used to determine the maximum window size"** ❌ Incorrect – The window size is negotiated during the TCP session but not directly part of the handshake.
- **"It uses a sliding window mechanism for flow control"** ❌ Incorrect – TCP does use a sliding window for flow control, but this occurs **after** the handshake.

Thus, the correct answer is:
"It involves the exchange of a sequence number, acknowledgment number, and a checksum"

Congestion Control

Q9

Congestion window of TCP connection be 64 KB when a timeout occurs. The RTT of connection is 80 milli sec and the maximum segment size used is 4KB. The time taken by TCP connection to get back to 64 KB congestion window is?

- 1100 milli sec
- 880 milli sec
- 960 milli sec
- 800 milli sec

Answer: 880 milliseconds

Given Data:

- **Initial congestion window (cwnd) after timeout** = **1 MSS** = **4 KB**
- **Target congestion window** = **64 KB**
- **RTT** = **80 milliseconds**
- **Maximum Segment Size (MSS)** = **4 KB**

Step 1: Slow Start (Exponential Growth)

In **slow start**, TCP **doubles the congestion window** every RTT until it reaches **32 KB**.

RTT	cwnd (KB)
0	4

1	8
2	16
3	32

At this point, slow start **stops**, and congestion avoidance begins.

Total time for slow start:

$3 \times 80 = 240$ ms

Step 2: Congestion Avoidance (Linear Growth)

- In congestion avoidance, **cwnd increases by 1 MSS (4 KB) per RTT**.
- We need to grow from **32 KB to 64 KB**, which requires **8 RTTs**.

Total time for congestion avoidance:

$8 \times 80 = 640$ ms

Final Calculation

Total time = **Slow Start Time + Congestion Avoidance Time**

$240 + 640 = 880$ ms

Correct Answer: 880 milliseconds

Q10

Which of the following mechanisms is implemented by TCP to avoid network congestion?

- Flow control
- Slow-start mechanism

- Sliding window
- All of the above

Answer: All of the above

TCP includes multiple mechanisms to **avoid network congestion**:

- **Flow Control** – Prevents overwhelming the receiver by managing data transmission speed.
- **Slow-Start Mechanism** – Gradually increases the congestion window (cwnd) to probe for available bandwidth.
- **Sliding Window** – Controls how many packets can be sent before receiving an acknowledgment.

Since **all** of these contribute to congestion control, the correct answer is:

"All of the above"

Security and Cryptography

Q11

Which of the following is the primary function of the TLS protocol in network communications?

- Ensuring packet delivery over unreliable networks with encryption at IP layer
- Encrypting data for secure communication between client and server at Socket Layer
- Adds flow management feature to a TCP connection at transport layer
- Translating IP addresses between networks by adding an extra tunnel header

Answer: Encrypting data for secure communication between client and server at Socket Layer

TLS (Transport Layer Security) is a cryptographic protocol that:

- **Encrypts communication** between client and server to prevent eavesdropping and tampering.
- **Operates at the Transport Layer (or between Transport and Application Layer)**, commonly used in HTTPS.
- **Ensures authentication, integrity, and confidentiality of data.**

Why the other options are incorrect?

❌ **"Ensuring packet delivery over unreliable networks with encryption at IP layer"** → Incorrect, as this describes IPsec, not TLS.
❌ **"Adds flow management feature to a TCP connection at transport layer"** → Incorrect, as flow control is handled by TCP itself, not TLS.
❌ **"Translating IP addresses between networks by adding an extra tunnel header"** → Incorrect, as this describes **NAT (Network Address Translation)**, not TLS.

Correct Answer: Encrypting data for secure communication between client and server at Socket Layer

Q12

Which of the following is true about IPsec?

- It secures the transport layer communication using public key encryption
- It operates at the link layer, providing end-to-end encryption
- It provides security for IP packets by authenticating and encrypting the payload
- It is primarily used for load balancing in VPNs

Answer: It provides security for IP packets by authenticating and encrypting the payload

IPsec (Internet Protocol Security):

- **Secures IP packets** by providing **encryption and authentication**.
- **Operates at Layer 3 (Network Layer)**, making it independent of transport protocols like TCP or UDP.
- **Uses ESP (Encapsulating Security Payload) and AH (Authentication Header)** for securing communications.
- **Commonly used for VPNs (Virtual Private Networks)** to encrypt data over insecure networks.

Why the other options are incorrect?

❌ **"It secures the transport layer communication using public key encryption"** → Incorrect, as IPsec works at the **network layer**, not the transport layer.
❌ **"It operates at the link layer, providing end-to-end encryption"** → Incorrect, as IPsec operates at the **network layer (Layer 3)**, not the **link layer (Layer 2)**.
❌ **"It is primarily used for load balancing in VPNs"** → Incorrect, as IPsec is used for **secure communication**, not load balancing.

Correct Answer: It provides security for IP packets by authenticating and encrypting the payload

Q13

In cloud networks, which protocol is commonly used for securing communication between different applications running on different VMs/Containers?

- IPSec
- SSL/TLS
- OpenVPN
- Secure Shell (SSH)

Answer: SSL/TLS

- **SSL/TLS** is the most commonly used protocol for securing communication **between applications running on different VMs or containers** in cloud environments.
- **IPSec** is mainly used for securing network-level communications (e.g., VPNs).
- **OpenVPN** is a VPN protocol, not specifically for inter-VM communication.
- **SSH** is used for secure remote access, not application-to-application security.

Correct Answer: SSL/TLS

Q14

Which of the following algorithms is an example of symmetric-key cryptography?

- RSA
- Diffie-Hellman
- AES
- ECDSA

Answer: AES

- **Symmetric-key cryptography** uses the **same key for both encryption and decryption**.
- **AES (Advanced Encryption Standard)** is a widely used symmetric encryption algorithm.
- **RSA, Diffie-Hellman, and ECDSA** are all **asymmetric cryptography** techniques.

✅ **Correct Answer: AES**

Q15

Which of the following is a feature of asymmetric-key cryptography?

- Same key is used for encryption and decryption
- It requires only a single key pair
- One key is used for encryption, and a different key is used for decryption
- It provides faster encryption than symmetric-key cryptography

Answer: One key is used for encryption, and a different key is used for decryption.

- In **asymmetric cryptography**, there are **two keys**:
  - A **public key** (used for encryption).
  - A **private key** (used for decryption).
- This makes asymmetric cryptography **useful for secure key exchange and authentication**.

Why the other options are incorrect?

❌ **"Same key is used for encryption and decryption"** → This describes **symmetric** encryption, not asymmetric.
❌ **"It requires only a single key pair"** → While asymmetric cryptography uses key pairs, this statement is misleading.
❌ **"It provides faster encryption than symmetric-key cryptography"** → Asymmetric encryption is **slower** than symmetric encryption due to computational complexity.

Correct Answer: One key is used for encryption, and a different key is used for decryption.

Q16

Company Xspace is designing a high-performance security chip for galactic communications using a proprietary protocol X. The benchmark used for testing protocol X requires 20 MD5 hashes, 30 SHA1 hashes, and 6 encryption operations. In the current chip, the latencies are:

MD5 hash: 5ns

SHA1 hash: 10ns

Encryption: 100ns

In the next generation of their chip, they have two options:

Option 1: Increase the hash performance by a factor of 2

Option 2: Increase the encryption performance by a factor of 1.6

Which option should they choose for higher performance and why?

- Option 1, because the hash operations take less time and increasing their performance by a factor of 2 will yield a greater performance improvement.
- Option 2, because encryption operations dominate the total time, and increasing encryption performance by a factor of 1.6 will reduce the overall time more significantly.
- Option 1, because MD5 and SHA1 hashes are more important in the benchmark than encryption
- Option 2, because the encryption operations are less time-consuming and will not provide significant benefits when improved

Correct Answer:
Option 2, because encryption operations dominate the total time, and increasing encryption performance by a factor of 1.6 will reduce the overall time more significantly.

We analyze the total processing time for each chip generation.

Current Performance:

Total time=$(20\times5)+(30\times10)+(6\times100)$ = 100 + 300 + 600 =1000 ns

Option 1: Doubling Hash Performance

- New MD5 latency = **5 ns / 2 = 2.5 ns**
- New SHA1 latency = **10 ns / 2 = 5 ns**
- Encryption remains the same (100 ns)

New total time:

$(20\times2.5)+(30\times5)+(6\times100)$ = 50 + 150 + 600 =800 ns

Option 2: Increasing Encryption Performance by 1.6×

- New encryption latency = **100 ns / 1.6 = 62.5 ns**
- Hash operations remain the same

New total time:

$(20\times5)+(30\times10)+(6\times62.5)$ = 100 + 300 + 375 = 775 ns

(775 ns vs. 800 ns, so Option 2 results in a lower processing time.)

Latency Calculation, Sliding Window Protocols, Throughput Calculations

Q17

A router is processing 10 million packets per second (Mpps). If each packet is 500 bytes, what is the throughput in Gigabits per second (Gbps)?

- 5 Gbps
- 4 Gbps
- 40 Gbps
- 0.5 Gbps

Answer: 40 Gbps

Given:

- **Packet rate = 10 million packets per second (Mpps)**
- **Packet size = 500 bytes**
- **1 byte = 8 bits**

Throughput calculation:

$$\text{Throughput} = \text{Packet Rate} \times \text{Packet Size} \times 8$$

$$= 10,000,000 \times 500 \times 8$$

$$= 40,000,000,000 \text{ bits per second} = 40Gbps$$

Correct Answer: 40 Gbps

Q18

Assuming the minimum packet size is 64 bytes, what is the maximum packet rate that can be obtained over a 100 Gbps link, factoring in the L1 overheads?

- 142 million packets per second
- 150 million packets per second
- 160 million packets per second
- 195 million packets per second

Correct Answer:
150 million packets per second
To calculate the maximum packet rate, we factor in L1 overheads and use the formula:

$$\text{Packet Rate} = \frac{\text{Link Speed (bits/sec)}}{\text{Packet Size (bits)}}$$

Given:

- Link speed = 100 Gbps = $\mathbf{10^{11}}$ **bits/sec**
- Minimum packet size = **64 bytes**
- L1 overhead:
  - 64-byte frame + 8-byte preamble + 12-byte interframe gap = **84 bytes**
  - 84 bytes = **84 × 8 = 672 bits**

$$\text{Max Packet Rate} = \frac{10^{11}\ \text{bits/sec}}{672\ \text{bits/packet}}$$

$=1.488\times10^{8}$ packets/sec=148.8 Mpps ≈150 Mpps

Q19

Which of the following can be a factor influencing latency in a network?

- Propagation delay
- Bandwidth of the link
- Queuing delay at routers
- All of the above

Answer: All of the above

Latency in a network is influenced by multiple factors:

1. **Propagation Delay** – The time taken for a signal to travel through the medium (depends on distance and speed of transmission).
2. **Bandwidth of the Link** – Affects how much data can be transmitted at once; lower bandwidth can cause buffering delays.
3. **Queuing Delay at Routers** – Packets may need to wait in a queue before being processed and forwarded.

Since all of these impact network latency, the correct answer is:

"All of the above"

Q20

In Go-Back-N ARQ, if the sender's window size is 4, which of the following is true?

- The sender can send up to 4 packets before waiting for acknowledgment
- The sender can send only 1 packet at a time
- The sender must wait for acknowledgment after sending the first packet
- The receiver can send acknowledgments out of order

Answer: The sender can send up to 4 packets before waiting for acknowledgment.

- In **Go-Back-N (GBN) ARQ**, the sender can send multiple packets (up to the window size) **without waiting for an acknowledgment**.
- If the sender's window size is **4**, it can **send 4 packets before requiring an ACK**.
- If a packet is lost, the sender **must retransmit all unacknowledged packets** starting from the lost one.

Correct Answer: The sender can send up to 4 packets before waiting for acknowledgment.

Q21

Consider 2 hosts X and Y connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is $2*10^8$ m/sec. Host X sends a file of 60,000 Bytes as one large message to host Y continuously. Let transmission and propagation delay be p milliseconds and q milliseconds respectively. The value of p+q is :

- 540
- 510
- 520
- 530

Answer: 530 milliseconds

Solution:

Step 1: Calculate Transmission Delay (t_p)

- **File size = 60,000 Bytes = 60,000×8 bits = 480,000 bits**
- **Link rate = 10^6 bits/sec**

$$\text{Transmission Delay} = \frac{\text{Total bits}}{\text{Transmission rate}} = \frac{480{,}000}{10^6} = 480 \text{ ms}$$

Step 2: Calculate Propagation Delay (t_d)

- **Distance = 10,000 km = 10^7 meters**
- **Propagation speed = 2×10^8 m/s**

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Propagation Speed}} = \frac{10^7}{2\times10^8} = 50 \text{ ms}$$

Final Calculation:

p+q=480+50=530 ms

✅ **Correct Answer: 530 milliseconds**

Q22

If a file of 10 GB is transferred over a 1 Gbps link, how much time will it take to complete the transfer?

- 80+ seconds
- 10+ seconds
- 80+ minutes
- 10+ minutes

Answer: 80+ seconds

Solution:

- **File size = 10 GB = 10×8 = 80 Gbits**
- **Link speed = 1 Gbps**

$$\text{Time required} = \frac{\text{Total bits to transfer}}{\text{Transfer rate}} = \frac{80 \text{ Gbits}}{1 \text{ Gbps}} = 80 \text{ seconds}$$

✅ **Correct Answer: 80+ seconds**

RDMA, RoCE, Infiniband

Q23

What does RDMA (Remote Direct Memory Access) enable in a network?

- High-speed encryption of data before transmission
- Direct memory-to-memory data transfer between computers without involving the CPU
- Error-free data transmission over unreliable networks
- Scaling of bandwidth by aggregating multiple links

Answer: Direct memory-to-memory data transfer between computers without involving the CPU

RDMA allows:

- **Direct data transfer from one computer's memory to another's** without CPU intervention.
- **Bypassing the operating system** to reduce overhead.
- **Low latency and high throughput**, making it ideal for HPC, cloud computing, and storage networks.

Why the other options are incorrect?

❌ **"High-speed encryption of data before transmission"** → RDMA is for efficient data transfer, but encryption is handled separately.
❌ **"Error-free data transmission over unreliable networks"** → RDMA does not inherently provide error correction; it relies on reliable transport layers.
❌ **"Scaling of bandwidth by aggregating multiple links"** → RDMA does not handle link aggregation; that is done by other networking techniques.

Correct Answer: Direct memory-to-memory data transfer between computers without involving the CPU

Q24

RoCEv2 leverages which combination of technologies to achieve RDMA over standard Ethernet?

- Fibre Channel and TCP
- Ethernet frames and iSCSI
- RDMA, UDP/IP, and Ethernet
- InfiniBand and UDP

Answer: RDMA, UDP/IP, and Ethernet

RoCEv2 (RDMA over Converged Ethernet v2) uses:

- **RDMA for direct memory access over the network.**
- **UDP/IP to enable RDMA over routable networks.**
- **Ethernet as the underlying transport technology.**

Why the other options are incorrect?

❌ **"Fibre Channel and TCP" → RoCEv2 does not use Fibre Channel or TCP. It uses UDP/IP instead of TCP.**
❌ **"Ethernet frames and iSCSI" → iSCSI is for block storage over IP, not related to RDMA.**
❌ **"InfiniBand and UDP" → RoCEv2 runs over Ethernet, not InfiniBand.**

Correct Answer: RDMA, UDP/IP, and Ethernet

Q25

Which of the following features makes InfiniBand particularly suitable for high-performance computing (HPC) environments?

- Support for low-latency and high-throughput communication
- Use of a single protocol for both networking and storage
- Broad compatibility with standard Ethernet hardware
- Integrated support for IP addressing

Answer: Support for low-latency and high-throughput communication

InfiniBand is widely used in HPC environments because:

- **Ultra-low latency** (~microseconds or lower).
- **High bandwidth**, supporting speeds beyond Ethernet.
- **Scalability for large HPC clusters**.

Why the other options are incorrect?

❌ **"Use of a single protocol for both networking and storage" →** InfiniBand does support networking and storage but uses multiple protocols

(e.g., IP over InfiniBand, SRP for storage).
❌ **"Broad compatibility with standard Ethernet hardware"** → InfiniBand uses its own specialized hardware, not standard Ethernet hardware.
❌ **"Integrated support for IP addressing"** → InfiniBand can support IP, but it is not its main feature; it uses its own addressing mechanism.

Correct Answer: Support for low-latency and high-throughput communication

Q26

Which of the following is NOT true about InfiniBand?

- It supports RDMA-based communication.
- It is widely used in High-Performance Computing (HPC).
- It has higher latency than traditional Ethernet.
- It can operate in different link speeds (e.g., SDR, DDR, QDR, EDR).

Answer: It has higher latency than traditional Ethernet.

- **InfiniBand is designed for ultra-low latency** and high throughput.
- It is commonly used in **HPC (High-Performance Computing)** clusters.
- It supports **RDMA-based communication**, which allows direct memory access without CPU involvement.
- It operates at **different link speeds** like SDR (Single Data Rate), DDR (Double Data Rate), QDR (Quad Data Rate), EDR (Enhanced Data Rate), etc.

Correct Answer: It has higher latency than traditional Ethernet. (This statement is false.)

IP addressing, Subnetting

Q27

Given the IP address 192.168.10.25/29, which of the following statements is correct regarding its subnet and broadcast address?

- The subnet address is 192.168.10.24 and the broadcast address is 192.168.10.31.
- The subnet address is 192.168.10.16 and the broadcast address is 192.168.10.23.
- The subnet address is 192.168.10.32 and the broadcast address is 192.168.10.39.
- The subnet address is 192.168.10.0 and the broadcast address is 192.168.10.7.

Answer: The subnet address is 192.168.10.24 and the broadcast address is 192.168.10.31.

Solution:

- **Given IP:** `192.168.10.25/29`
- **Subnet Mask:** `/29` corresponds to `255.255.255.248`, meaning 3 host bits.
- **Subnet size:** $2^3=8$ addresses per subnet.
- **Finding the subnet range:**
  - Nearest multiple of **8** just below `25` is **24**, so subnet starts at `192.168.10.24`.
  - The broadcast address is the last address in the subnet: `192.168.10.31`.

Correct Answer: The subnet address is 192.168.10.24 and the broadcast address is 192.168.10.31.

TCP Flow Control

Q28

Which of the following is the function of TCP flow control?

- To prevent congestion by regulating packet transmission rates
- To control the size of the receiver’s buffer and prevent buffer overflow
- To establish a connection between sender and receiver
- To manage retransmissions of lost packets

Answer: To control the size of the receiver’s buffer and prevent buffer overflow

Explanation:

- **TCP Flow Control** ensures that the sender does not overwhelm the receiver’s buffer.
- The **receiver advertises a window size (rwnd)** that tells the sender how much data it can handle.
- This prevents **buffer overflow** at the receiver’s end.

Why are the other options incorrect?

❌ **"To prevent congestion by regulating packet transmission rates"** → This is **Congestion Control**, not Flow Control.
❌ **"To establish a connection between sender and receiver"** → This is part of the **TCP three-way handshake**, not Flow Control.
❌ **"To manage retransmissions of lost packets"** → This is handled by **TCP reliability mechanisms**, not Flow Control.

Correct Answer: To control the size of the receiver’s buffer and prevent buffer overflow

NAT

Q29

Which of the following best describes the purpose of NAT (Network Address Translation)?

- It translates domain names into IP addresses
- It modifies source or destination IP addresses of packets to route them through firewalls
- It encrypts data for secure communication
- It splits large packets into smaller ones for transmission

Correct Answer:
"It modifies source or destination IP addresses of packets to route them through firewalls."

Explanation:
NAT is primarily used to modify IP addresses in packet headers to enable multiple devices in a private network to share a single public IP address. This is common in home and corporate networks, where private IP addresses (e.g., 192.168.x.x) need translation to a public IP when communicating over the internet. It also adds a layer of security by masking internal network addresses from external entities

Timeout Calculation

Q30

A sender measures SampleRTT values for a connection and computes EstimatedRTT and DevRTT accordingly. Given the following values:

Initial EstimatedRTT = 200 ms

Initial DevRTT = 20 ms

SampleRTT at time t = 220 ms

α=0.125 β=0.25

What will be the new Timeout Interval using the TCP timeout calculation formula?

- 260 ms
- 280 ms
- 300 ms
- 320 ms

We calculate the new **EstimatedRTT, DevRTT,** and **TimeoutInterval** using the formulas:

- **EstimatedRTT = (1 - α) * EstimatedRTT + α * SampleRTT**
- **DevRTT = (1 - β) * DevRTT + β * |SampleRTT - EstimatedRTT|**
- **TimeoutInterval = EstimatedRTT + 4 * DevRTT**

Given:

- Initial **EstimatedRTT** = 200 ms
- Initial **DevRTT** = 20 ms
- SampleRTT = 220 ms
- α = 0.125, β = 0.25

Step 1: Compute New EstimatedRTT

$$\text{EstimatedRTT} = (1 - 0.125) \times 200 + 0.125 \times 220$$

$$= 0.875 \times 200 + 0.125 \times 220$$

$$= 175 + 27.5 = 202.5 \text{ ms}$$

Step 2: Compute New DevRTT

$$\text{DevRTT} = (1 - 0.25) \times 20 + 0.25 \times |220 - 202.5|$$

$$= 0.75 \times 20 + 0.25 \times 17.5$$

$$= 15 + 4.375 = 19.375 \text{ ms}$$

Step 3: Compute Timeout Interval

$$\text{TimeoutInterval} = 202.5 + 4 \times 19.375$$

$$= 202.5 + 77.5 = 280 \text{ ms}$$

Correct Answer:
B) 280 ms